Optimizer Hints: Overview

Optimizer hints:

- Influence optimizer decisions
- Example:

```
SELECT /*+ INDEX(e empfirstname_idx) skewed col */ *
FROM employees e
WHERE first_name='David'
```

- HINTS SHOULD ONLY BE USED AS A LAST RESORT.
- When you use a hint, it is good practice to also add a comment about that hint.

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Optimizer Hints: Overview

Hints enable you to influence decisions made by the optimizer. Hints provide a mechanism to direct the optimizer to select a certain query execution plan based on the specific criteria.

For example, you might know that a certain index is more selective for certain queries. Based on this information, you might be able to select a more efficient execution plan than the plan that the optimizer recommends. In such a case, use hints to force the optimizer to use the optimal execution plan. This is illustrated in the slide example where you force the optimizer to use the `EMPFIRSTNAME_IDX` index to retrieve the data. As you can see, you can use comments in a SQL statement to pass instructions to the optimizer.

The plus sign (+) causes the system to interpret the comment as a list of hints. The plus sign must follow immediately after the comment delimiter. No space is permitted.

Hints should be used sparingly, and only after you have collected statistics on the relevant tables and evaluated the optimizer plan without hints using the `EXPLAIN PLAN` statement. Changing database conditions as well as query performance enhancements in subsequent releases can have a significant impact on how hints in your code affect performance.

In addition, the use of hints involves extra code that must be managed, checked, and controlled.

Types of Hints

Single-table hints	Specified on one table or view
Multitable hints	Specify more than one table or view
Query block hints	Operate on a single query block
Statement hints	Apply to the entire SQL statement

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Types of Hints

Single-table: Single-table hints are specified on one table or view. `INDEX` and `USE_NL` are examples of single-table hints.

Multitable: Multitable hints are like single-table hints, except that the hint can specify one or more tables or views. `LEADING` is an example of a multitable hint.

Query block: Query block hints operate on single query blocks. `STAR_TRANSFORMATION` and `UNNEST` are examples of query block hints.

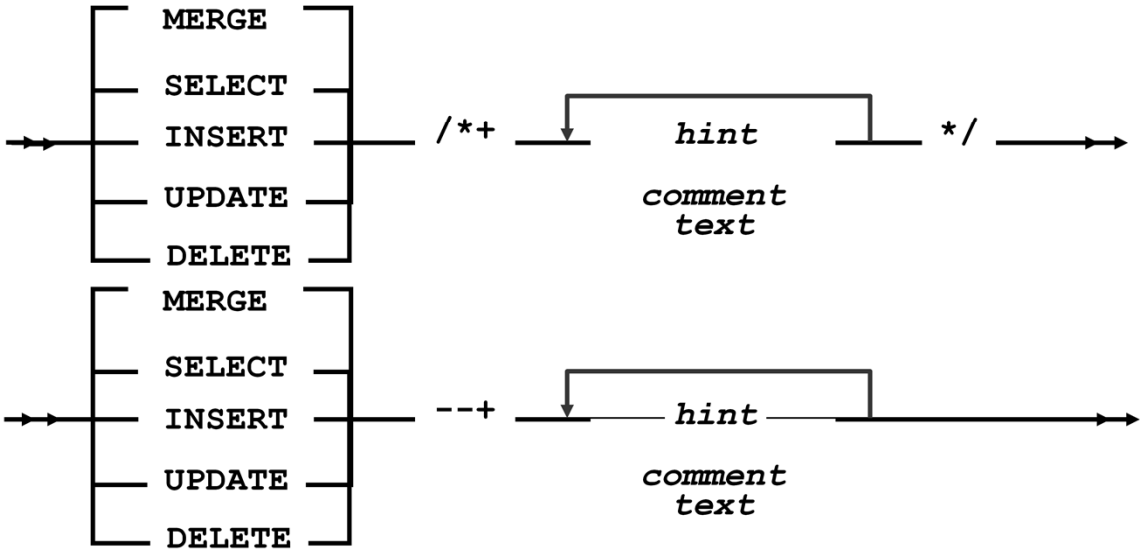
Statement: Statement hints apply to the entire SQL statement. `ALL_ROWS` is an example of a statement hint.

Note: `USE_NL(table1 table2)` is not considered a multitable hint because it is actually a shortcut for `USE_NL(table1)` and `USE_NL(table2)`.

Specifying Hints

Hints apply to the optimization of only one statement block:

- A self-contained DML statement against a table
- A top-level DML or a subquery



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Specifying Hints

Hints apply to the optimization of only the block of the statement in which they appear. A statement block is:

- A simple `MERGE`, `SELECT`, `INSERT`, `UPDATE`, or `DELETE` statement
- A parent statement or a subquery of a complex statement
- A part of a compound query using set operators (`UNION`, `MINUS`, `INTERSECT`)

For example, a compound query consisting of two component queries combined by the `UNION` operator has two blocks, one for each component query. For this reason, hints in the first component query apply only to its optimization, not to the optimization of the second component query.

Optimizer Hint Syntax

Enclose hints within the comments of a SQL statement. You can use either style of comment. The hint delimiter (`+`) must come immediately after the comment delimiter. If you separate them by a space, the optimizer does not recognize that the comment contains hints.

Rules for Hints

- Place hints immediately after the first SQL keyword of a statement block.
- Each statement block can have only one hint comment, but it can contain multiple hints.
- Hints apply to only the statement block in which they appear.
- If a statement uses aliases, hints must reference the aliases rather than the table names.
- The optimizer ignores hints specified incorrectly without raising errors.

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Rules for Hints

- You must place the hint comment immediately after the first keyword (`MERGE`, `SELECT`, `INSERT`, `DELETE`, or `UPDATE`) of a SQL statement block.
- A statement block can have only one comment containing hints, but it can contain many hints inside that comment separated by spaces.
- Hints apply to only the statement block in which they appear and override instance- or session-level parameters.
- If a SQL statement uses aliases, hints must reference the aliases rather than the table names.

The Oracle optimizer ignores incorrectly specified hints. However, be aware of the following situations:

- You never get an error message.
- Other (correctly) specified hints in the same comment are considered.
- The Oracle optimizer also ignores combinations of conflicting hints.

Hint Recommendations

- Use hints carefully because they imply a high-maintenance load.
- Be aware of the performance impact of hard-coded hints when they become less valid.

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Hint Recommendations

- Use hints as a last remedy when tuning SQL statements.
- Hints may prevent the optimizer from using better execution plans.
- Hints may become less valid (or even invalid) when the database structure or contents change.

Optimizer Hint Syntax: Example

```
UPDATE /*+ INDEX(p PRODUCTS_PROD_CAT_IX)*/
products p
SET   p.prod_min_price =
        (SELECT
         (pr.prod_list_price*.95)
         FROM products pr
         WHERE p.prod_id = pr.prod_id)
WHERE p.prod_category = 'Men'
AND   p.prod_status = 'available, on stock'
/
```


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Optimizer Hint Syntax: Example

The slide shows an example with a hint that advises the cost-based optimizer (CBO) to use the index. The execution plan is as follows:

```
Execution Plan
----------------------------------------------------------
   0      UPDATE STATEMENT Optimizer=ALL_ROWS (Cost=3 …)
   1    0   UPDATE OF 'PRODUCTS'
   2    1     TABLE ACCESS (BY INDEX ROWID) OF 'PRODUCTS' (TABLE)
(Cost…)
   3    2      INDEX (RANGE SCAN) OF 'PRODUCTS_PROD_CAT_IX' (INDEX)
                  (cost…)
   4    1     TABLE ACCESS (BY INDEX ROWID) OF 'PRODUCTS' (TABLE)
(Cost…)
   5    4       INDEX (UNIQUE SCAN) OF 'PRODUCTS_PK' (INDEX (UNIQUE))
                 (Cost=0 …)
```

The hint shown in the example works only if an index called `PRODUCTS_PROD_CAT_IX` exists on the `PRODUCTS` table in the `PROD_CATEGORY` column.